CHITRESH SUNIL DHAKE
Pune, Maharashtra | 7517426083 | dhakechitresh@gmail.com
LinkedIn http://www.linkedin.com/in/chitresh-dhake-652298381

PROFESSIONAL SUMMARY

Manufacturing and Industrial Engineer with 2+ years of experience in automotive and aerospace wire harness assembly, pre-series runs and engineering changes (ECN). Delivered a **44% line capacity increase**, **₹58.4L in cumulative annual labor** and material cost savings, and **₹1.6 Cr in CAPEX avoidance** through fixture re-engineering. Hands-on experience working on **prototype and series builds for OEM programs** including **Tesla, Tata, VW, and Mahindra**. Practical background conducting video and stopwatch time studies, Yamazumi line balancing, PFMEA inputs, and BOM verification against ERP/SAP production data.

TECHNICAL SKILLS

- **Industrial Engineering & Lean:** Time & Motion Study, Line Balancing, Yamazumi Balancing, Workstation Ergonomics (REBA), 5S, Kaizen.
- **NPI & Shop-Floor Execution:** Pre-Series & Prototype Runs, Workstation Layouts, Standard Operating Procedures (SOP), Work Instructions (WI), Line Takt Alignment.
- **Engineering Change Support**: ECN/PCN verification, PFMEA participation, Control Plan and PPAP support, 5-Why analysis, FAI assistance, and adherence to IPC guidelines.
- **Systems & Data Verification:** Drawing & Work Order Verification, Wire Length Calculation (WLC) Validation, Shop-Floor ECN Implementation Support, Basic Exposure to Power BI, and MS Excel.
- **Design & CAD Tools:** SolidWorks, AutoCAD, CATIA (Basic Knowledge), Engineering Drawings and GD&T.

PROFESSIONAL EXPERIENCE

Manufacturing Engineer (Industrial Engineer)
Aptiv (EDS) - Pune | [June 2025 –Present]

- Coordinated the rollout of 50+ ECNs with zero line downtime, validating component availability across 100% of affected BOM line items prior to changeover.
- Handled pre-series pilot assembly runs across 4 major OEM programs (Tesla, Tata, Mahindra HV, and Hyundai HV), maintaining 100% on-time dispatch for all prototype lots.
- Deployed Volkswagen ADAS harness routing changes and KSK assembly work standards across rear electrical harness stations, ensuring full adherence to OEM launch milestones.
- Supported OS Audits, Internal Audits, and Customer Audits by preparing required records and documents.
- Authored and updated 25+ assembly SOPs, work element sheets, and visual inspection aids to standardize new takt times and routing sequences after engineering changes.
- Contributed Industrial Engineering inputs during cross-functional PFMEA reviews, auditing station-level risk items and validating run-at-rate packaging data for PPAP submissions.
- Analyzed line bottleneck and station cycle time trends using Power BI and advanced excel reports to identify balancing losses and reallocate operator work elements

Industrial Engineer Trainee
GKN Aerospace (Fokker Elmo) | [June 2024 –May 2025]

- Conducted video-based motion analysis and stopwatch time studies across aerospace wiring cells, breaking down tasks into VA/NVA elements on aerospace standard observation sheets to set baseline cycle times.
- Assisted in FAI documentation, harness verification, and quality checks in accordance with AS9100D and customer requirements.
- Participated in PFMEA activities for new process implementation, including Laser Marking Fixture development, and supported identification of potential process risks and control measures.
- Supported NCR investigations and participated in 5-Why root cause analysis for production-related issues.
- Evaluated high-repetition assembly benches using REBA postural scoring, recommending bin and bench layout adjustments that cut severe bending motions by 80%.
- Received the GKN Kaizen Event Trophy and 3 Kaizen of the Month recognitions for shop-floor productivity improvements.

PROJECT

- **Mahindra XUV Assembly Line Debottlenecking (*Aptiv EDS*):** Addressed output constraints on the main assembly track by conducting VSM and Yamazumi cycle audits across 32 individual stations. Reallocated connector plug-in tasks, raising monthly line capacity from **9,700 to 14,000 harness sets (44% volume uplift)**.
- **Final Assembly Manpower Optimization (*Aptiv EDS*):** Rebalanced workstation loads using time-study logs, decommissioning 2 navigation stations and saving 2 operators per shift **(₹22.4L annual recurring savings**).

- **Wire Length Standardization & Trim-Waste Reduction (*Aptiv EDS*):** Validated drawing cut lengths against physical routing fixtures using WLC tools, correcting master dimension tables and cutting **₹36L in annual wire scrap**.
- **Laser Marking Fixture Re-Engineering (*GKN Aerospace*):** Designed a multi-cavity indexing holding fixture in SolidWorks to batch-mark 30 backshells per cycle on the ACI laser cell, **increasing machine output by 90%** and avoiding **₹1.6 Cr in CAPEX**.
- **Aerospace Braiding Process Cycle Time Reduction (*GKN Aerospace*):** Evaluated protective wire braiding through Time & motion studies, reducing cycle time from **9.3 to 5 hours (46% reduction)** and converting the station from a two-operator to a single-operator task.
- **Smart Modular Rooftop Farming Structure Optimization (*Academic Project*):** Analyzed and optimized a modular structural frame for weight reduction; co-authored research paper published in IEEE https://ieeexplore.ieee.org/document/11132144.

EDUCATION

- **B.Tech** – Mechanical Engineering
  JSPM Rajarshi Shahu College of Engineering, Pune | **CGPA: 9.3** |2025
- **Diploma** – Mechanical Engineering
  Government Polytechnic Ambad | **CGPA: 8.62**|2022.

CERTIFICATIONS & ACHIEVEMENTS

- Six Sigma Yellow Belt (SSYB) – 6 Sigma Study
- IPC Wire Harness Assembly Standards Awareness (Aerospace)
- AS9100D Quality Management Standards Awareness
- Lean Manufacturing & 7 QC Tools Internal Training (Aptiv & GKN)
- Certified AutoCAD Draftsman | Practical Training in SolidWorks & CATIA
- Advanced Machining Training (CIPET) | Locomotive In-Plant Training (Indian Railways)